# Lesson 13C Gravitational Potential Energy

## Gravitational Potential Energy

Gravitational potential energy is a property of a system of two particles, not of an individual particle. It cannot be assigned separately to each particle in the pair.

However, in many practical situations—such as a ball near Earth—we refer to "the gravitational potential energy of the ball." This simplification works because the change in energy manifests almost entirely as kinetic energy of the smaller object (the ball), due to the large mass difference between the objects.

The gravitational potential energy $U$ of a two-particle system is given by:

$$U = -\frac{GMm}{r}$$

- $M$ and $m$: masses of the two particles
- $r$: distance between their centers
- $G$: gravitational constant

This potential energy is **negative**, and it **approaches zero** as $r \to \infty$.
For a system of multiple particles, the total gravitational potential energy is the **sum over all distinct pairs** of particles.

## Conservative Nature of Gravitational Force

Gravitational force is a **conservative force**:

- The work done by gravity depends only on the starting and ending positions, not on the path taken.
- Therefore, the change in gravitational potential energy also depends only on these positions.

In fact, Newton's law of gravitation can be derived by taking the derivative of the potential energy with respect to $r$.

# Escape Speed

The **escape speed** is the minimum speed an object must have at the surface of a celestial body in order to move infinitely far away without being pulled back by gravity.

To escape, the object must reach infinity with **zero kinetic energy remaining**. That is:

$$K = 0 \quad \text{and} \quad U = 0 \quad \text{at} \quad r \to \infty$$

We apply conservation of mechanical energy. At the surface of the planet or star, the object has:

- Kinetic energy: $K = \frac{1}{2}mv^2$
- Gravitational potential energy: $U = -\frac{GMm}{R}$

At infinity:

- $K = 0$
- $U = 0$

Using conservation of energy:

$$K_{\text{initial}} + U_{\text{initial}} = K_{\text{final}} + U_{\text{final}}$$

$$\frac{1}{2}mv^2 - \frac{GMm}{R} = 0$$

Solving for $v$:

$$\frac{1}{2}mv^2 = \frac{GMm}{R}$$

$$v^2 = \frac{2GM}{R}$$

$$v_{\text{escape}} = \sqrt{\frac{2GM}{R}}$$

Where:

- $v_{\text{escape}}$: escape speed
- $G$: gravitational constant
- $M$: mass of the planet or star
- $R$: distance from the center of the mass (typically the radius of the planet/star)

This is the **minimum speed** required to leave the object's gravitational influence without any additional propulsion.

## Black Holes

A **black hole** is the final stage of a star that has collapsed under its own gravity.

As the star's radius decreases while retaining its mass, the gravitational force at its surface increases dramatically. The escape speed eventually becomes greater than the speed of light:

$$v_{\text{escape}} > c$$

Since **not even light can escape**, the object appears completely **black**.

## Black Holes and Accretion Disks

Even though a black hole emits no light, matter falling into it can become extremely hot, emitting light and radiation before it crosses the event horizon.

In a **binary star system**, if one star is a black hole and the other is a normal star, material from the normal star may be pulled into the black hole. This process forms an **accretion disk**, which glows brightly and makes the black hole observable indirectly.

Black Holes: Crash Course Astronomy #33

Brian Greene on the first-ever image of a black hole from th...